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Prediction of cardiovascular diseases based on GBDT+LR

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Currently, there are over 300 million patients with cardiovascular diseases in China. With the acceleration of population aging, the impact of cardiovascular diseases is becoming increasingly severe. Accurately and efficiently predicting the potential risks of cardiovascular disease is crucial for preventing its progression and maintaining public cardiovascular health. This article uses a combination of gradient-boosting decision trees (GBDT) and logistic regression (LR) to predict the probability of cardiovascular disease risk. To address the weak feature combination ability of LR in handling nonlinear data, a cardiovascular disease prediction model was established by integrating GBDT with LR by using the predicted results of GBDT as new features instead of the original ones and inputting them into the LR model. Using the UCI cardiovascular disease dataset, we conduct experimental comparisons between the proposed model and other common disease classification algorithms such as logistic regression (LR), random forest (RF), and support vector machine (SVM). The experimental results show that GBDT+LR outperforms other models in multiple evaluation indicators such as accuracy, precision, specificity, F1 value, MCC, AUC, and AUPR. The cardiovascular disease prediction model using the GBDT+LR algorithm has the best prediction performance. This article builds a front-end and back-end separated cardiovascular disease analysis and prediction platform based on the Spark Big data framework and Vue+SpringBoot framework, which realizes predicting cardiovascular disease risk probability.

Keywords Spark, Cardiovascular disease, Prediction, Machine learning, GBDT, LR, GBDT+LR

Abbreviations

LR	Logistic regression
RF	Random forest
SVM	Support vector machine
GBDT	Gradient boosting decision trees
AUC	Area under the ROC curve
AUPR	Area under the precision-recall curve
MCC	Matthews correlation coefficient
CVD	Cardiovascular disease
UCI	University of California, Irvine
F1	F1-score

Cardiovascular disease is the leading cause of death worldwide[1], and one of the three major diseases recognized by the World Health Organization[2]. According to data from the Morbidity and Mortality Weekly Report released by the U.S. Centers for Disease Control and Prevention, approximately 3.45 million people died in the United States in 2021[3]. Among them, cardiovascular disease accounted for the highest number of deaths, exceeding those caused by cancer and COVID-19. With nearly 700,000 deaths, it was responsible for approximately one-fifth of all fatalities in the United States that year. In China, cardiovascular diseases also pose a significant threat to public health. The China Cardiovascular Health and Disease Report 2022 reveals that the incidence rate and mortality of cardiovascular diseases in China is still increasing. The number of patients with hypertension is about 245 million. Cardiovascular diseases are the number one killer of the Chinese population. Two out of every five disease deaths nationwide are due to cardiovascular diseases, and the mortality rate is even higher than that of tumors[4]. Given the widespread impact and severity of cardiovascular disease, early detection of potential risks and effective preventive measures remain critical challenges in the medical field.

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A large number of studies have shown that cardiovascular disease can be artificially prevented and controlled through early intervention. In the early stage of the disease, some physiological indicators in the human body will change. If these changes can be detected timely and accurately, then the cardiovascular disease can be predicted, and the occurrence of cardiovascular disease can be significantly reduced or even avoided[5]. As early as 1967, Kannel et al. proposed the risk function of integrating various pathogenic factors for disease prediction, and earlier introduced logistic regression into medical research, verifying that the simple model established by using these characteristics can effectively predict cardiovascular disease[6]. Since then, Wang et al.[7] have included seven factors, including age, body mass index (BMI), smoking, drinking, blood pressure, blood glucose and blood lipid status, into the logistic regression model, achieving a good effect of cardiovascular disease risk prediction.

In recent years, machine learning technology has been widely used in the field of cardiovascular disease prediction and achieved good results because of its powerful data processing ability[8,9]. Compared with the standard statistical model, machine learning (ML) algorithm is better at capturing the complex interactions and nonlinear relationships between variables and results, and can be used to find the abnormalities of blood pressure, blood glucose, cholesterol and other physiological indicators in patients with cardiovascular disease, so as to improve the prediction accuracy[5]. For example, Nawaz et al. used Gradient Descent Optimization, Support Vector Machine, K-Nearest Neighbor, ANN and Random Forest to predict cardiovascular diseases[10]. The results show that the prediction model based on gradient descent optimization achieves the best effect in performance evaluation, and the prediction accuracy rate reaches 98.54%. Zhang also proposed the enhanced feature level deep convolutional neural network for the prediction of cardiovascular disease, and the F1 value of its model reached 0.9516[11]. An and Huang proposed to use Bi-RNN neural network to capture the characteristics of the electronic medical record data of patients with cardiovascular disease, which improved the effect of cardiovascular risk prediction model[12]. Cai combined SVM algorithm with DS theory and proposed a SVM-DS prediction model to predict cardiovascular disease[13]. Du etc used XGBoost algorithm to build a prediction model of coronary heart disease risk, and the AUC value reached 0.967[14]. In addition, Sivakannan et al. used the SHAP method combined with GBDT for feature selection, and the experimental results showed that the model performed well in the task of cardiovascular disease prediction[9].

Although there have been many researches on machine learning in the prediction of cardiovascular and cerebrovascular diseases, literature retrieval shows that GBDT+LR model has not been applied in this field. The GBDT+LR model was first proposed in 2014. Its principle is to use the GBDT model to automatically carry out feature combination and screening, and then use the newly generated discrete feature vector as the input of the LR model, so as to comprehensively use a variety of different features to generate more comprehensive prediction results. This model was initially used by Facebook as an ad recommendation model to predict the possibility of users clicking on ads. Once proposed, the GBDT+LR model has been widely used in industry[15–19]. However, in the existing literature, the application of GBDT+LR model in the prediction of cardiovascular disease has not been found.

Based on this, this paper intends to introduce GBDT+LR model into the field of cardiovascular disease prediction, explore the feasibility and effectiveness of GBDT+LR model in cardiovascular disease prediction, and develop a cardiovascular disease analysis and prediction platform based on this model. The platform can analyse the physiological indexes of users, and realize the prediction and early warning of cardiovascular disease data. The establishment of the cardiovascular analysis and prediction platform helps to achieve rapid and low-cost cardiovascular disease detection, facilitate the tracking and analysis of cardiovascular health for users, and help medical staff realize the long-term monitoring of cardiovascular health, so as to normalize the prediction of cardiovascular disease.

The main contributions of this article are as follows:

- The first application of the GBDT+LR algorithm in the field of cardiovascular disease prediction, demonstrating its potential for improving prediction accuracy.
- Build a machine-learning disease prediction model and use it in a Spark-based cardiovascular disease prediction platform. Random Forest, Support Vector Machine, LR, GBDT, and GBDT+LR algorithms were used to build cardiovascular and cerebrovascular disease prediction models respectively, and were applied to the UCI data set. The accuracy rates reached 71.5%, 69.3%, 71.4%, 72.4%, and 78.3% respectively. After comprehensively comparing the precision, accuracy, sensitivity, specificity, F1 value, etc. of different models, the GBDT+LR model was finally selected as the final cardiovascular disease prediction model, embedded in a Spark-based Java web application for cardiovascular disease prediction.
- Designed and developed a cardiovascular disease analysis and prediction platform based on the GBDT+LR model. This article uses the Spark big data framework to train disease prediction models, uses Java language, and combines the Vue + Spring Boot framework to build a front-end and back-end separated cardiovascular disease analysis and prediction system. The GBDT+LR-based cardiovascular disease prediction model is embedded in the system, achieving functions such as user management, disease prediction, data analysis, and display. The platform can be used to assess potential cardiovascular disease risks, predict cardiovascular risks, and take timely preventive or therapeutic measures to reduce the harm caused by cardiovascular diseases.

The rest of this paper is organized as follows. In "Materials and methods" section, the Cardiovascular disease prediction process and data used in this study are briefly described, and then data pre-processing and descriptive statistical analysis are presented, after which machine learning models and evaluation metrics used are introduced. The main results are shown in "Results" section, discussions are included in "Discussion" section, followed by the conclusion in "Conclusion" section.

Materials and methods

Method description

To solve the problem of the weak feature combination ability of the LR model for nonlinear data, the GBDT algorithm is introduced to fully utilize the advantages of the ensemble learning algorithm in processing nonlinear data. A cardiovascular disease prediction model (GBDT+LR) was established by integrating GBDT with LR by using the predicted results of GBDT as new features instead of the original ones and inputting them into the LR model. Specifically, the cardiovascular disease prediction process designed in this article is shown in Fig. 1:

Figure 1 shows the flowchart of cardiovascular disease prediction. In the entire process of cardiovascular disease prediction, the raw data is preprocessed and trained using the GBDT model to obtain a new feature combination. Then, the new feature combination is combined with the original features to train the LR model and obtain the final classification result.

Data description and preprocessing

This article uses the Cardiovascular Disease open-source dataset published by UCI for cardiovascular disease prediction. The dataset can be downloaded from: https://www.kaggle.com/datasets/sulianova/cardiovascular-disease-dataset. This dataset includes data from approximately 70000 normal individuals and cardiovascular patients. There are a total of 12 indicators in the dataset. Among them, the first 11 indicators are cardiovascular disease risk factors, including the patient's basic information(Objective: age, gender, height, weight), the patient's measurement indicators (Examination: systolic blood pressure, diastolic blood pressure, cholesterol level, blood glucose level), and the patient's behavioral factors (Subjective: smoking, drinking, activity). The 12th indicator is the diagnosis result of cardiovascular disease. It is a label in the sample and is used as a prediction object in the experiment. The meanings of each field in the dataset are shown in Table 1.

Before building a model using data, it is necessary to preprocess the data to obtain data that can be directly used for analysis. This article performed the following preprocessing work on the original data.

- Handling of missing data;
  After exploring the dataset, it is found that all attributes in the dataset have no missing values, so no additional processing is required.
- Handling of outliers.
  The basic statistics of the dataset are shown in Table 2:

As shown in Table 2, some attribute values are negative or exceed the normal range (e.g., the minimum diastolic blood pressure is negative). To eliminate the impact of abnormal data on model training, we refer to the standard range of each attribute and remove outliers accordingly.

Based on box plot visualization, this study adopts the double interquartile range (IQR) as the criterion for outlier detection. Specifically, for each numerical attribute, IQR is calculated as the difference between the 75th percentile (Q3) and the 25th percentile (Q1). Any data point exceeding Q1 - step × IQR or Q3 + step × IQR is considered an outlier, where step controls the strictness of detection (default value: 1.5).

Taking the weight attribute as an example, its box plot is shown in Fig. 2. The presence of numerous outliers is evident, appearing as points outside the box range. These values significantly deviate from the normal

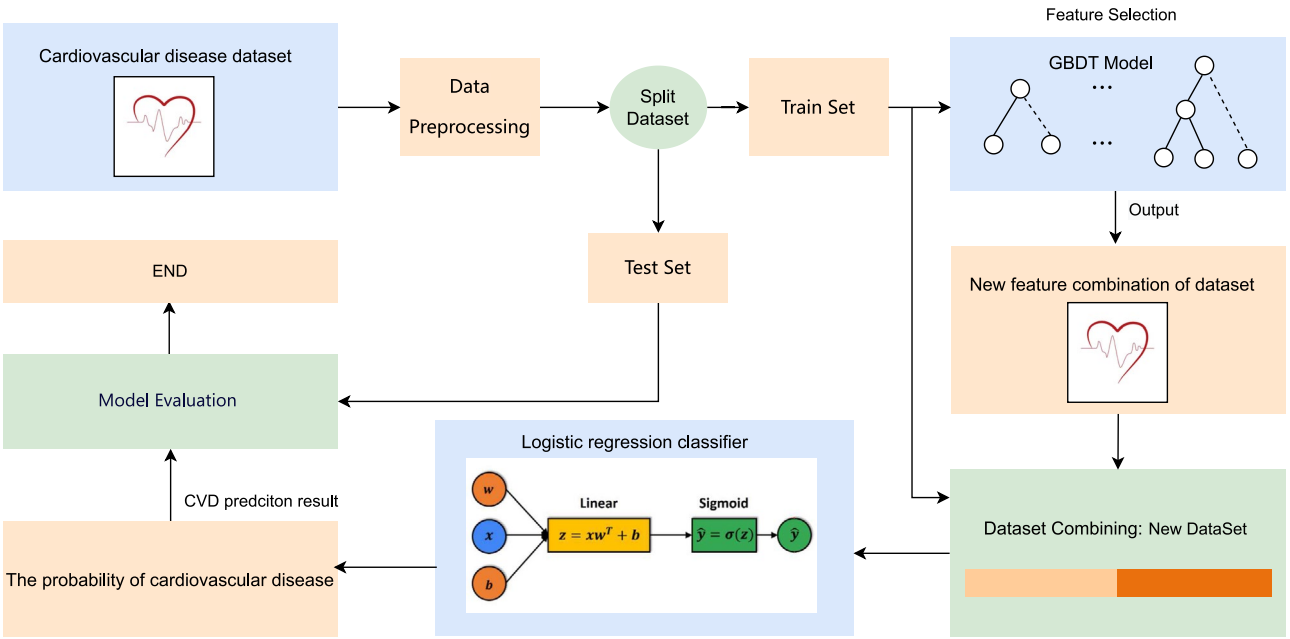


Fig. 1. GBDT+LR cardiovascular disease prediction flow chart.

Field name	Field meaning
Age	Age, unit: years
Gender	Gender| 1:male, 2:female
Height	Height, unit: cm
Weight	Weight, unit: kg
Ap_hi	Systolic blood pressure
Ap_low	Diastolic blood pressure
Cholesterol	cholesterol | 1: normal, 2: above normal, 3: well above normal
Gluc	Glucose| 1: normal, 2: above normal, 3: well above normal
Smoke	Smoking or not | 1: smoking, 2: non-smoking
Active	Physical activity| 1: less, 2: more
Alco	Whether to drink alcohol| 1: drinking, 2: not drinking
Cardio	Presence or absence of cardiovascular disease | 0: non-diseased, 1: deseased

Table 1. Meaning of each field in the dataset.

	Age	Gender	Height	Weight	ap_hi	ap_lo
Count	70,000	70,000	70,000	70,000	70,000	70,000
Mean	19,468.866	1.350	164.359	74.206	128.817	96.630
std	2467.252	0.477	8.210	14.396	154.011	188.473
min	10,798.000	1.000	55.000	10.000	-150.000	-70.000
25%	17,664.000	1.000	159.000	65.000	120.000	80.000
50%	19,703.000	1.000	165.000	72.000	120.000	80.000
75%	21,327.000	2.000	170.000	82.000	140.000	90.000
max	23,713.000	2.000	250.000	200.000	16,020.000	11,000.000

Table 2. Partial basic statistics of the dataset.

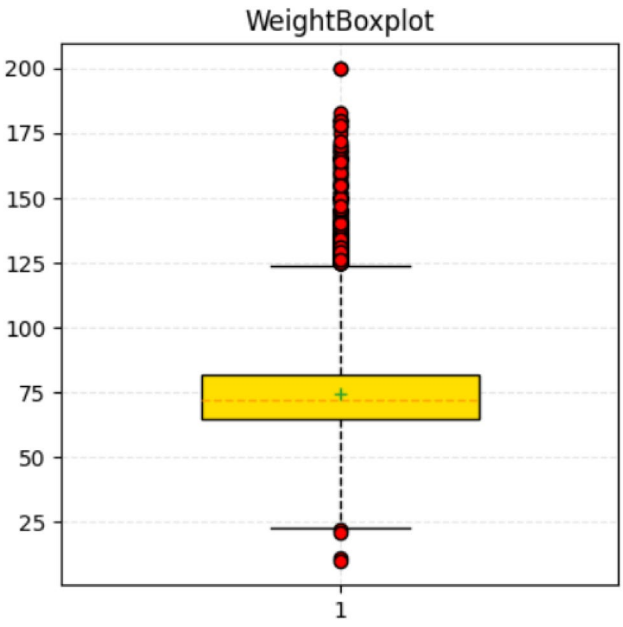


Fig. 2. Boxplot for weight.

distribution. To enhance dataset robustness and improve model generalization, we remove all outliers based on the above method. The post-processing box plot (Fig. 3) shows that the remaining data now falls within the planned standard range, completing the data-cleaning process. This step effectively mitigates the impact of extreme values, improving both the accuracy and stability of model training.

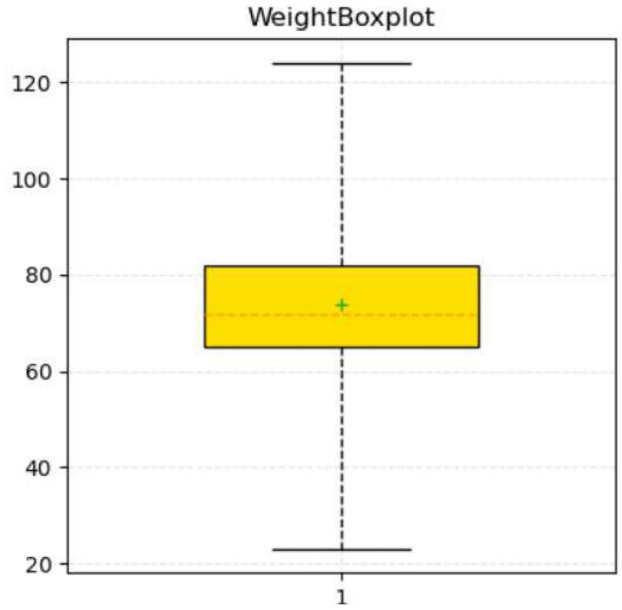


Fig. 3. Boxplot for weight data after cleaning.

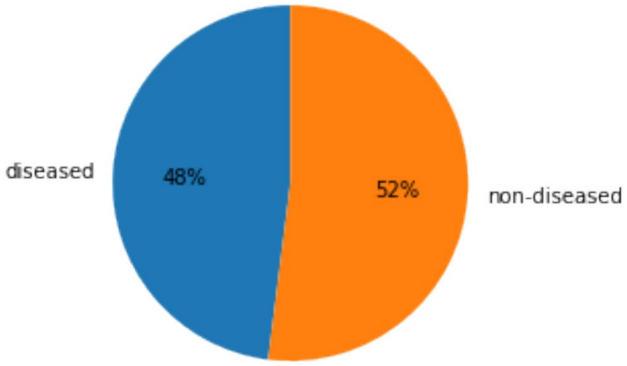


Fig. 4. Disease category ratio chart.

After cleaning, the dataset contains 68,283 records, including 32,776 cases in the diseased category and 35,507 in the non-diseased category. The proportion of each category is illustrated in Fig. 4

To assess potential covariance issues, we generated a heatmap of correlation coefficients for all features, as shown in Fig. 5. The results showed that no absolute correlation exceeds 0.8. Studies suggest that when $|r| > 0.8$, multicollinearity may be a concern, requiring reduction of dimension or removal of features [20,21]. Therefore, our data do not exhibit severe multicollinearity, and high feature correlation is unlikely to cause overfitting in GBDT.

Prediction models

Logic regression model

The logistic regression model is a machine learning model with a simple form and good interpretability [22]. As a commonly used classification and prediction algorithm, logistic regression models are widely used in multiple application fields, such as garbage information discrimination, risk identification in the financial field, and disease prediction. It is a linear regression classification model that can solve problems such as regression, binary classification, and multi-classification [16]. The core idea of logistic regression is to map the output of linear regression to the probability space through the logarithmic probability function, thereby achieving classification. It is a commonly used medical data analysis method [15]. The principle of disease prediction is based on the linear regression algorithm, using the Sigmoid function to transform the predicted probability value, mapping the dependent variable to [0, 1], and then classifying it into different categories according to the set threshold to achieve disease prediction.

The process of the logistic regression algorithm is as follows:

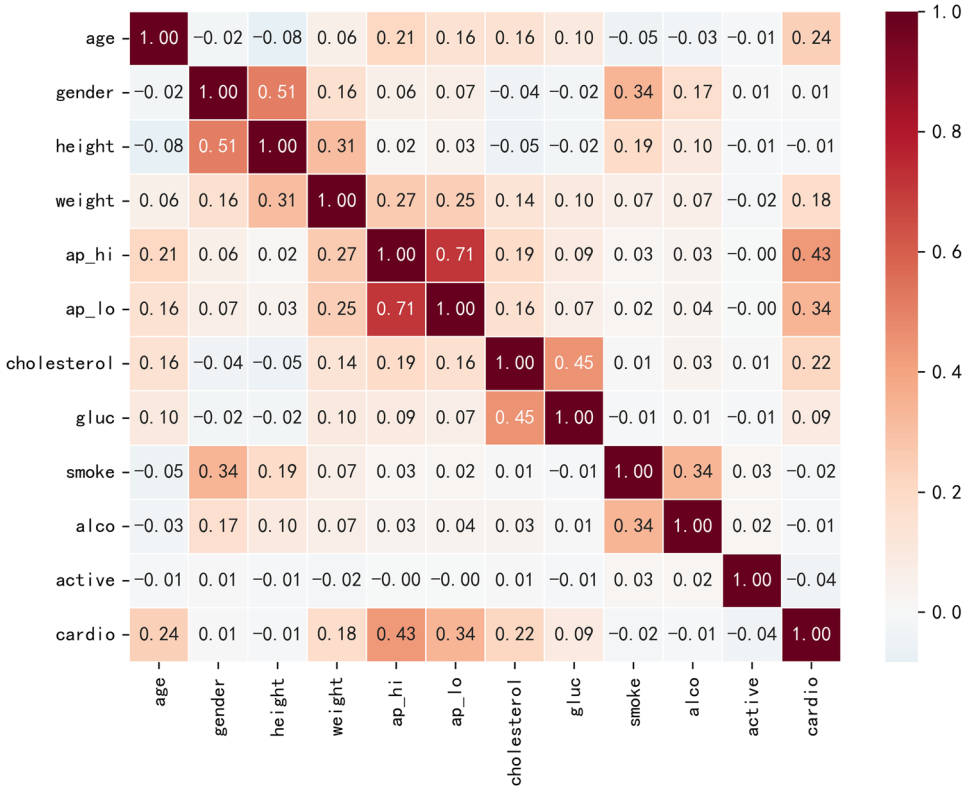


Fig. 5. Correlation heatmap of features.

1. Assume x is a set of continuous random variables that follow a Logistic distribution. The cumulative distribution function and density function of x are:

$$F(x) = P(X \leq x) = \frac{1}{1 + e^{-(x-u)/\gamma}} \tag{1}$$

$$f(x) = \frac{e^{-(x-\mu)/\gamma}}{\gamma(1 + e^{-(x-\mu)/\gamma})^2} \tag{2}$$

2. The expression for the Logistic function is:

$$g(z) = \frac{1}{1 + e^{-z}} \tag{3}$$

3. Set the training set $T = (x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$, where $x_i \in R^n, y_i \in \{0, 1\}$. Suppose$P(Y = 1|x) = \pi(x)$, for parameter w, the log-likelihood function after taking the logarithm is:

$$L(w) = \sum_{i=1}^{n} [y_i(w \cdot x_i + b) - log(1 + e^{w \cdot x_i + b})] \tag{4}$$

4. Use gradient descent to estimate the value of w and obtain the final prediction probability.

$$P(x) = p(y = 1|x) = \frac{e^{wx+w_0}}{1 + e^{wx+w_0}} \tag{5}$$

The main advantages of the logistic regression algorithm are: (1) When implementing classification, the model training speed is fast, the resource consumption is low, and it is suitable for scenarios with massive data. (2) The implementation of the algorithm is relatively simple and very suitable for binary classification problems. (3) Multilinear problems can be solved by combining L1 and L2 regularization. But at the same time, the logistic regression algorithm also has some shortcomings: (1) When the feature space is large, the performance of logistic

regression is poor. (2) Logistic regression cannot be used to solve nonlinear problems, and feature conversion is usually required. (3) The accuracy of the model is not high.

Gradient boosting decision tree model

The gradient boosting decision tree(GBDT) method is an ensemble learning algorithm for additive models based on the Ada Boost framework proposed by Friedman in 1999[23]. It belongs to a type of iterative decision tree model. As one of the ensemble algorithms, this method combines a series of weak prediction models and has shown good predictive performance in many fields[24,25]. It can solve regression problems and be applied in classification scenarios and is widely used in the industry[24–33].

Because each iteration of GBDT requires fitting continuous gradient values, GBDT uses CART regression trees as base learners for both classification and regression[28,34]. It classifies and analyzes the residual between the regression tree and the target value. It gradually improves the algorithm to continuously reduce the residual so that the calculated value gradually approaches the target value. Finally, the results obtained by each weak classifier are weighted and summed to obtain the final classifier[35]. The GBDT model fits each tree's residual data, which helps prevent overfitting, and has a strong generalization ability. It can be combined with the logistic regression model for predicting cardiovascular diseases.

The main idea of the GBDT algorithm is to train a new learner in the gradient direction to reduce the learning error rate of the previous learner, and the new learner is iteratively generated based on the previous learner[36]. The algorithm process is as follows:

1. Take the training set $(x_1, y_1), (x_2, y_2), \cdots, (x_N, y_N)$ and estimate the model parameter γ that minimizes the loss to initialize the weak classifier:

$$f_0(x) = \arg\min_{\gamma} \sum_{i=1}^{N} L(y_i, \gamma) \tag{6}$$

2. Let M be the number of iterations and for m=1, 2,..., M, perform the following steps:

   ① Calculate the negative gradient of samples i=1,2,..., N one by one, and use it as approximate residual:

$$r_{mi} = -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f(x)=f_{m-1}(x)} \tag{7}$$

   ② Use $x_i, r_m i (i = 1, 2, \cdots, n)$ to fit a regression tree and obtain the leaf node domain $R_m j$ of the m-th tree, which is a tree composed of J leaf nodes.

   ③ For the leaf node region $j = 1, 2, \cdots, J_m$, calculate the best-fit value.

$$c_{m,j} = \underbrace{\operatorname{argmin}}_{c} \sum_{x_i \in R_{mj}} L\left(y_i, f_{t-1}\left(x_i\right) + c\right) \tag{8}$$

   ④ Update classifier

$$F_m(x) = F_{m-1}(x) + \sum_{j=1}^{J_m} c_{m,j} I\left(x \in R_{m,j}\right) \tag{9}$$

In the formula, $I\left(x \in R_{mj}\right) = \begin{cases} 1, x \in R_{mj} \\ 0, x \notin R_{mj} \end{cases}$

3. Obtain the final classification tree:

$$F_M(x) = F_0(x) + \sum_{m=1}^{M} \sum_{j=1}^{\mathrm{J}} c_{m,j} I\left(x \in R_{m,j}\right) \tag{10}$$

Advantages of the GBDT algorithm: (1) Compared with logistic regression, the prediction accuracy of this model is high. (2) It can handle nonlinear data and is suitable for low-dimensional data. (3) Supports multiple data types, supporting discrete values and continuous values.

The main shortcomings of the GBDT algorithm are: due to the dependency relationship between weak learners, it is difficult to train data in parallel, and it has high requirements for model parameters, resulting in long training time. Moreover, the model is not suitable for high-dimensional sparse data.

GBDT+LR model

In 2014, Facebook proposed a method that utilizes GBDT to automatically filter and combine features to generate new discrete feature vectors, and then use this feature vector as the input of the LR model to generate the final prediction result. This is the famous GBDT+LR model[37]. The GBDT+LR model uses GBDT to automatically combine and filter features and then uses LR to achieve target fitting. The entire process from data input, feature cross-combination, filtering, and model prediction output is automatically completed by the model. It is no longer necessary to manually combine and filter features, truly achieving end-to-end prediction[38].

The GBDT+LR model process is shown in Fig. 6:

The specific steps for constructing new features in the GBDT+LR model are as follows:

1. The GBDT model uses the input initial training set for training and constructs multiple decision trees to form a strong classifier.
2. When the trained GBDT model makes predictions, it does not output the probability of classification. Instead, it uses the position of the node corresponding to the predicted value of each tree in the model as a new feature to extract the feature value and construct a new feature.

The process of constructing new features is shown in Fig. 7:

Assuming that the GBDT algorithm generates two decision trees called f_{m-1} and f_m during the model training process, each tree has 5 leaf nodes, where the number 1 represents that the training sample is predicted to fall on this node through the decision tree. So for the decision tree f_{m-1}, one-hot encoding can be used to represent the prediction results as [0, 1, 0, 0, 0]. Assuming that the GBDT model has M iterations and all weak classifiers have a total of N leaf nodes, each of the Z original sample data will be transformed into a y-dimensional sparse vector, where M elements are 1 and the rest are 0, a new training set of $Z \times y \times N$ dimensions will be formed[12,28].

3. Encode the newly constructed features to obtain the label vector M of the sample data, so that the outputs of all samples are combined into a sparse matrix.

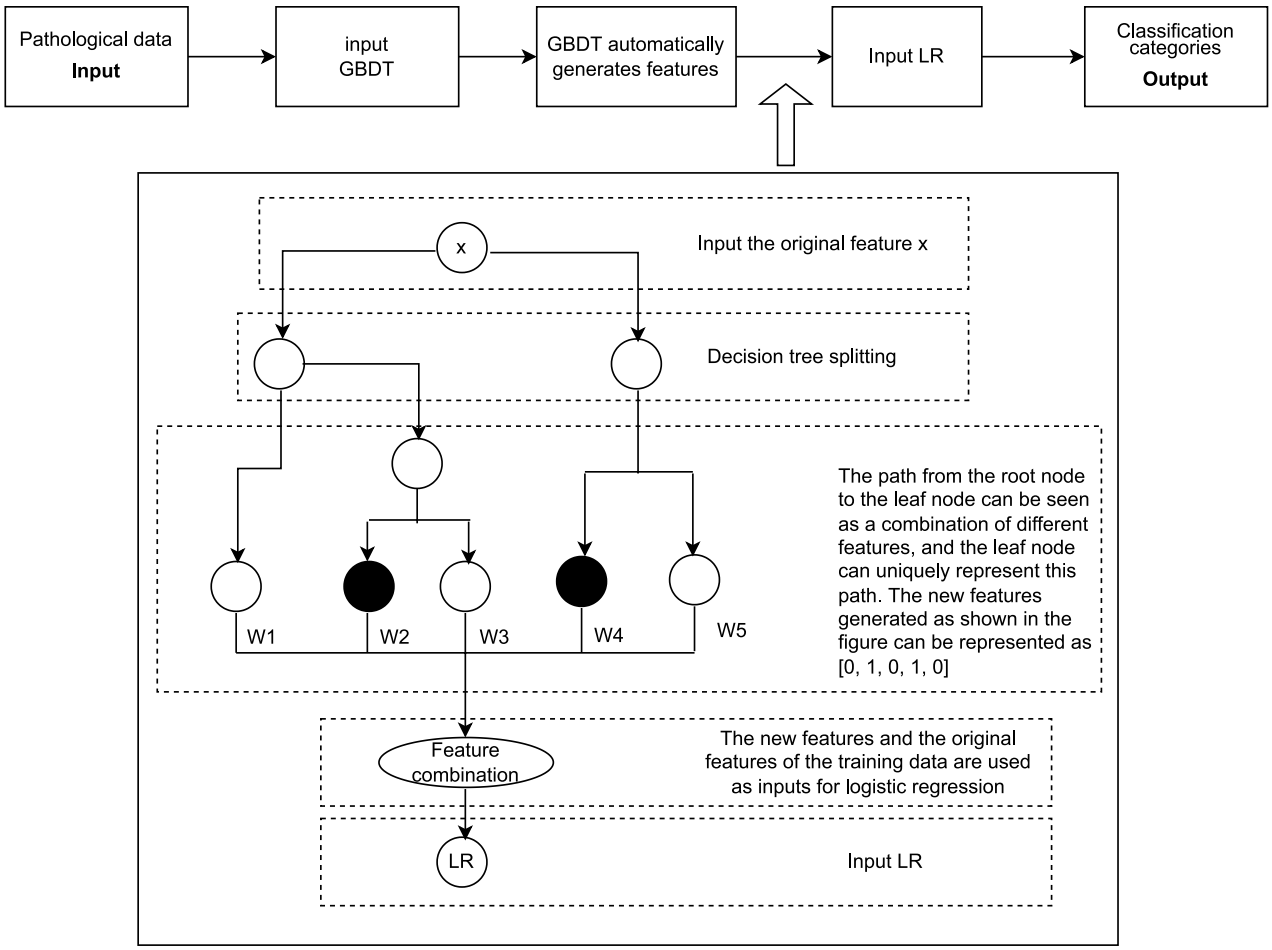


Fig. 6. GBDT+LR feature construction.

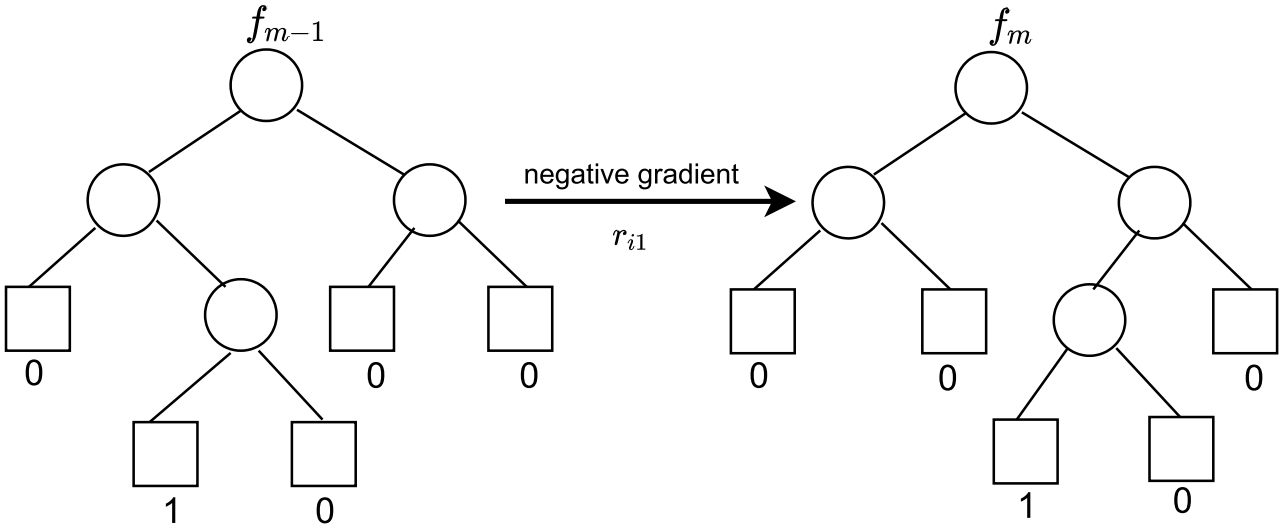


Fig. 7. GBDT constructs new feature process.

Actual class	Predicted class	
	Diseased	Non-diseased
Diseased	TP	FN
Non-diseased	FP	TN

Table 3. Confusion matrix structure.

4. Train the LR model using the labeled vector M as new training data.
5. Obtain the predicted category of the sample.

The process of GBDT+LR algorithm is shown in Algorithm 1:

Input: $T = \{(x_1, y_1), (x_2, y_2), \cdots, (x_N, y_N)\}$ //Training sample data
Output: $P(x)$
1: $f_0(x) = 0$ //Base classifier initialization
2: $L(y, f(x)) = (y - f(x))^2$ //Define loss function
3: **for** m in $(1, M)$: **do**
4: $r_{mi} = -\left[\frac{\partial L(y_i, f(x_i))}{\partial f(x_i)}\right]_{f(x)=f_{m-1}(x)}$
5: $R_{mj}, j = 1, 2, \cdots, J\ M = \{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)\}$
6: **end for**
7: $L(w) = \sum_{i=1}^{n} \left[y_i (w \cdot x_i + b) - \log\left(1 + e^{w \cdot x_i + b}\right)\right]$
8: $P(x) = p(y = 1|x) = \frac{e^{wx+w_0}}{1+e^{wx+w_0}}$
9: **return** $P(x)$

Algorithm 1. GBDT+LR.

The GBDT+LR model combines the advantages of GBDT and logistic regression (LR) to effectively solve the shortcomings of the LR model in dealing with nonlinear relationships. GBDT can automatically discover and combine features, making up for the shortcomings of manual feature engineering and shortening the model training cycle. Although GBDT+LR is superior to the single LR model in prediction performance, its computational overhead is large, especially when processing large-scale data sets, the training time and memory consumption increase significantly[37]. However, in scenarios that require high-precision predictions (such as medical and financial), the superior performance of the GBDT+LR model makes it a viable choice. By optimizing model parameters (such as reducing the number of trees and controlling the depth of the tree), the computational overhead can be reduced to a certain extent and the training efficiency can be improved[37,39].

Evaluation metrics

This article selects the most commonly used confusion matrix as the evaluation indicator to compare the effectiveness of various models. The confusion matrix structure for cardiovascular disease prediction is shown in Table 3:

The confusion matrix shown in the table divides the classification results into four parts:

1. TP (True Positive): Actual diseased and predicted result is also diseased.
2. FN (True Negative): Actual diseased but predicted not to be affected.
3. FP (False Positive): Actual not affected but predicted to be affected.
4. TN (False Negative): Actual non-diseased and the predicted result is also non-diseased.

Based on the above concepts, the confusion matrix extends to more accurate evaluation indicators:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{11}$$

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

$$Specificity = \frac{TN}{FP + TN} \tag{13}$$

$$Sensitivity(recall) = \frac{TP}{TP + FN} \tag{14}$$

$$F_1 = 2 \times \frac{precision \times recall}{precision + recall} \tag{15}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN}} \tag{16}$$

In addition, this article also uses two other commonly used classification evaluation indicators, AUC and AUPR, whose calculation formulas are as follows:

$$AUC = \frac{\sum_{i \in positiveClass} rank_i - \frac{M*(M+1)}{2}}{M \times N} \tag{17}$$

$$AUPR = \sum_{k=1}^{n-1} (Recall_{K+1} - Recall_k) \cdot Precision_k \tag{18}$$

Accuracy refers to the percentage of correctly identified samples in the model predicted results to the total number of samples. At the same time, precision (also known as precision) refers to the percentage of samples predicted by the model as diseased samples compared to the total number of diseased samples. It refers to the ratio of correctly identified diseased samples in all predicted data[40]. Specificity and sensitivity can supplement the issue of incomplete representation of model performance by accuracy. The specificity describes the proportion of non-diseased samples identified to all non-diseased samples, that is, the ability to correctly identify non-diseased individuals. The lower the specificity, the higher the misdiagnosis rate; Sensitivity refers to the ability to accurately detect patients who do indeed have this disease, and is the ratio of people who test positive for the disease to those who have it. The higher the sensitivity, the lower the missed diagnosis rate[41]. The full name of F1 is the F1-score, which is the harmonic mean of accuracy and sensitivity. The larger the value, the better the performance of the model, and vice versa. MCC (Matthews coefficient) is an indicator used to evaluate the performance of binary classification models, especially suitable for imbalanced datasets. It considers the impact of the number and proportion of positive and negative samples on the classifier's performance comprehensively, with a value between [−1,1]. The closer the MCC value is to 1, the better the performance of the classifier[42]. AUC (Area Under the ROC Curve) refers to the probability that the model ranks a random positive example more highly than a random negative example. One of the advantages of AUC is its insensitivity to the selection of classification thresholds, so it can comprehensively evaluate the classifier's ability to rank positive and negative samples. The closer the value of AUC is to 1, the better the performance of the classifier[42]. AUPR (Area under the precision-recall curve) calculates and averages the accuracy and recall rates under different thresholds to obtain a comprehensive performance evaluation value. Compared to AUC, AUPR focuses more on the predictive ability of positive examples. The higher the value of AUPR, the better the model's performance in predicting positive examples[43]. In disease prediction, combining the indicators of actual scenarios for analysis, multiple indicators are used to comprehensively evaluate the model.

Results

In this experiment, to evaluate the performance of the GBDT+LR model in predicting cardiovascular diseases, four machine learning models were selected for comparative analysis, including LR (Logistic Regression), GBDT (Gradient Boosting Decision Tree), RF (Random Forest), and SVM (Support Vector Machine). Grid search and five-fold cross-validation were used for model optimization, and the selected classifier performance evaluation metrics indicators were precision, accuracy, sensitivity, specificity, and F1 score. The algorithm parameter settings are shown in Table 4 , and the experimental results are shown in Table 5.

From the accuracy, the highest prediction accuracy of the GBDT+LR classifier is 73.7%, with a single GBDT classifier having a prediction accuracy of 73.1%, and the single LR classifier has a prediction accuracy of 72%, both of which are lower than the prediction accuracy of the GBDT+LR classifier. The precision rate represents the accuracy of the model, and it is evident from the five models that the GBDT+LR classifier has a much

Algorithm	Parameter	Description	Value
LR	setMaxIter	Maximum iterations	30
	setRegParam	Regularization parameter (L2)	0.001
GBDT	setMaxIter	Maximum iterations	40
	setStepSize	Learning rate	0.001
RF	setNumTrees	Number of trees in the forest	10
	setMaxDepth	Maximum tree depth	3
	setSeed	Random seed for reproducibility	123
SVM	setMaxIter	Maximum iterations	30
	setRegParam	Regularization parameter (L2)	0.0
GBDT+LR	setGBTMaxIter	Maximum iterations for GBDT	10
	setLRMaxIter	Maximum iterations for Logistic Regression	30
	setRegParam	Regularization parameter (L2)	0.01
	setElasticNetParam	ElasticNet mixing parameter	0.5
	setMaxDepth	Maximum depth of the tree	6

Table 4. Algorithm parameters settings.

Model name	Precision	Accuracy	Sensitivity	Specificity	F1-score	MCC	AUC	AUPR
LR	0.720	0.732	**0.683**	0.756	0.707	0.441	0.771	0.820
GBDT	0.731	0.755	0.675	0.786	0.713	0.464	0.792	0.839
RF	0.721	0.742	0.666	0.774	0.702	0.443	0.781	0.818
SVM	0.704	0.737	0.622	0.783	0.675	0.411	0.766	0.811
GBDT+LR	**0.737**	**0.761**	0.682	**0.791**	**0.719**	**0.476**	**0.802**	**0.840**

Table 5. Performance comparison of different models.

higher prediction accuracy than the other four classifiers, indicating that the GBDT+LR classifier has the best classification performance. Except for the slightly lower sensitivity, the GBDT+LR model has the highest score in accuracy, precision, specificity, F1-score, MCC, AUC, and AUPR, which indicates that the model performs best in this task.

From the above table, it can be seen that the predictive performance of the GBDT+LR model is generally superior to the other four types of models. Compared to random forests, support vector machines, single logistic regression, and single gradient-boosting decision trees, there are improvements in accuracy, precision, specificity, F1 value, MCC, AUC, AUPR, etc. The experimental results indicate that the GBDT+LR fusion model used in this article performs well in predicting cardiovascular diseases.

The functional module diagram of the platform developed by this study is shown in Fig. 8:

Once the user has registered and logged in, they can input their data on the prediction page. The system will then transfer these data to the trained disease prediction model and initiate the Spark task to predict and obtain the specified prediction value. The obtained prediction value will be returned to the front end for display. Users can view their prediction records on a dedicated page, which is shown in Fig. 9.

Discussion

In this study, we used the GBDT+LR fusion algorithm on the UCI public dataset Cardiovascular disease to improve the prediction performance of cardiovascular diseases. Compared with commonly used random forests, support vector machines, and machine learning algorithms such as single GBDT and LR, the performance of the GBDT+LR algorithm is superior to other algorithm models in terms of precision, accuracy, sensitivity, specificity, and F1 value.

Compared with other traditional models, the GBDT part in GBDT+LR gives the model a natural advantage in discovering multiple discriminative features and feature combinations. The path of the decision tree can be directly used as the input feature for LR, eliminating the need for manual feature search and feature combination. The entire process of data input, feature cross combination, filtering, and model prediction output is automatically completed by the model, truly achieving end-to-end prediction. Since its proposal, the GBDT+LR model has been widely applied in the industry[15–19]. Wang constructed a GBDT-LR regression fusion model based on big data technology for precision marketing in the pharmaceutical economy. The effectiveness of the GBDT-LR model was demonstrated by comparing it with the purchasing behavior method and knowledge graph method[15]. Zhang et al. applied the GBDT+LR model to personal credit risk assessment[17], and the experimental results showed that the GBDT+LR model had better classification performance than models such as LR, DT, RF, and GBDT. Zeng et al. applied the GBDT-LR fusion algorithm to a fetal distress prediction model and compared it with models such as LR, SVM, BP neural network, backpropagation neural network, GBDT, and XGBoost. They concluded that GBDT-LR is superior to the above models[18]. However, according to the literature review, the GBDT+LR model

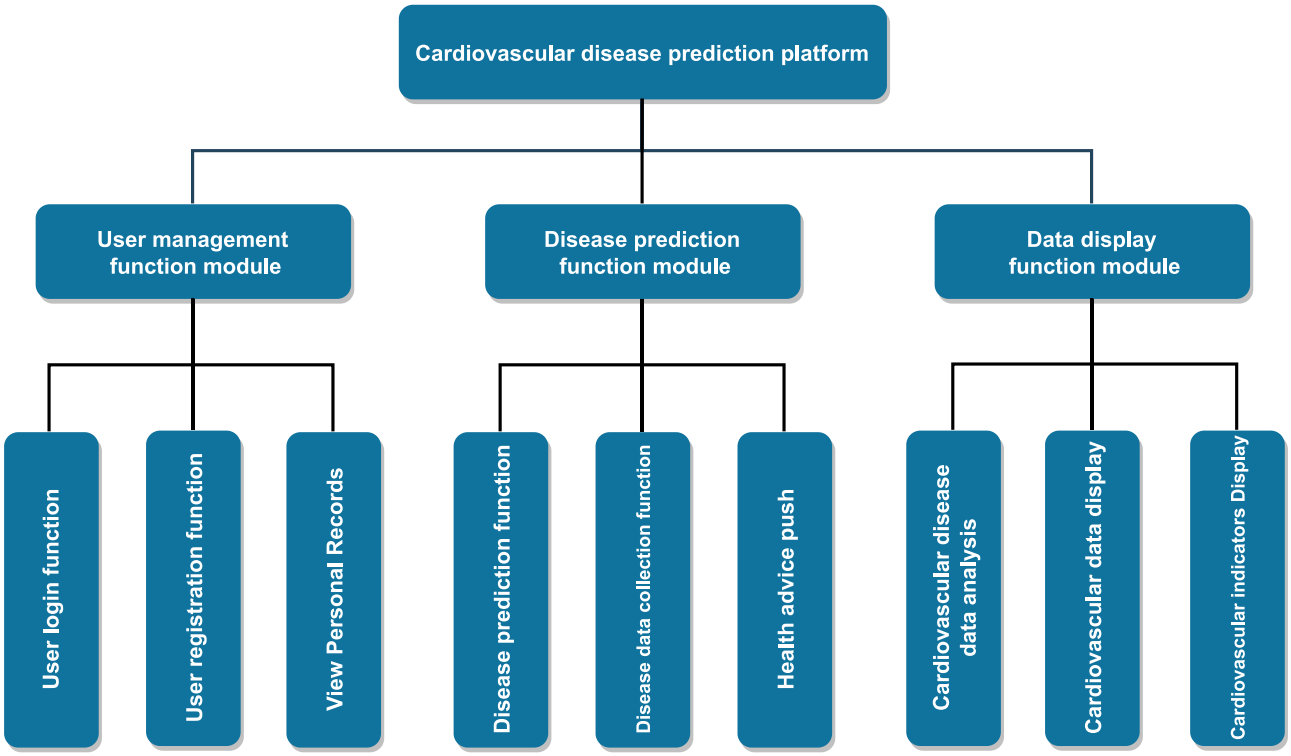


Fig. 8. Functional module diagram of cardiovascular disease prediction platform.

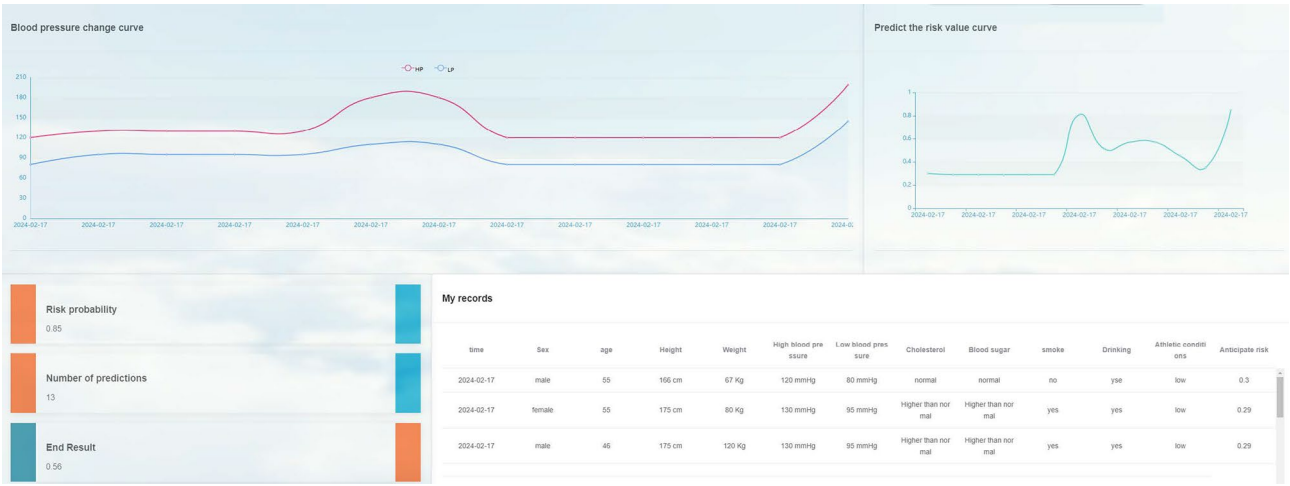


Fig. 9. Personal prediction record view page.

has not been used for predicting cardiovascular diseases. This article applies the GBDT+LR model to the field of cardiovascular disease prediction and explores the effect of the GBDT+LR model in predicting cardiovascular diseases. Consistent with the research of GBDT+LR in other fields, GBDT and LR can complement each other in predicting cardiovascular disease. GBDT can capture nonlinear relationships and complex interactions, while LR can handle linear relationships and reduce bias. The combination of the two can effectively utilize the nonlinear fitting ability of GBDT and the bias correction ability of LR. Compared with traditional machine learning models such as single GBDT and LR, GBDT+LR can better handle complex disease prediction problems and improve the accuracy of cardiovascular disease prediction. Compared with the references mentioned above, this article adopts more comprehensive evaluation indicators. However, the accuracy (0.800) of using the GBDT+LR model for predicting cardiovascular disease in this article is lower than that in references[17](0.9822) and[18](0.896). The possible reason is that the dataset used in reference[17] is LendingClub, a publicly available dataset with 148 rich features. Based on this, the author carefully selected 18 feature variables as input variables using the Wrapper feature selection method to train the model. Yet the UCI cardiovascular disease prediction dataset used in this

article only has 11 feature variables except for the label variables, which are not rich enough. In the study in reference[18], the GBDT-LR fusion model may exhibit overfitting due to insufficient data volume (1958 valid samples, including 1795 normal samples and 163 abnormal samples) as described by the author. In the follow-up work, we will further utilize more feature-rich cardiovascular disease prediction datasets and compare GBDT-LR with more traditional machine learning models to further explore the performance of the GBDT-LR model.

Based on analyzing and comparing GBDT+LR with other prediction algorithms, this article uses the GBDB+LR cardiovascular disease prediction model to build a cardiovascular disease analysis and prediction platform, analyzes various physical indicators of users, and realizes the prediction of cardiovascular disease data to warn users of potential cardiovascular disease risks early. The construction of this cardiovascular analysis and prediction platform helps to achieve fast and low-cost detection of cardiovascular diseases, facilitating the tracking and analysis of cardiovascular health for users, assisting medical staff in achieving long-term monitoring of cardiovascular health, and making cardiovascular disease prediction normalized.

Conclusion

This article constructs a cardiovascular disease prediction model using the GBDT+LR fusion method. The method first utilizes the GBDT algorithm to generate combined features from the original data. These combined features are then integrated with the original features before being input into the LR model for training, resulting in the final GBDT-LR model. The experimental results show that compared to existing single algorithm models such as random forest, support vector machine, LR, and GBDT, the GBDT-LR fusion algorithm model significantly enhances the prediction accuracy for cardiovascular disease and can assist doctors in achieving long-term cardiovascular disease risk monitoring, and make more effective assessment of the condition.

By combining GBDT for feature extraction and selection, the GBDT+LR model uses LR to fit the target. The entire process does not require manual intervention, truly achieving end-to-end prediction. As far as we know, this is the first time that the GBDT+LR algorithm has been used for cardiovascular disease prediction. Meanwhile, this article uses the fusion algorithm model of GBDT+LR, with the help of Spark big data computing technology and Vue+Spring Boot development framework, to build a cardiovascular disease prediction platform based on the B/S architecture with front-end and back-end separation. This platform plays a crucial role in helping users understand their cardiovascular health, prompting them to take preventive measures, and ultimately reducing the harm caused by cardiovascular diseases. The B/S-based architecture also improves the scalability and maintainability of the entire platform.

However, the platform built in this article has not yet been interconnected with other external data and devices. It is hoped that the prediction work in this article can be applied to multiple fields and devices in subsequent work. For example, cardiovascular disease data is shared with other disease data to explore the correlation between diseases. At the same time, it can achieve the data interconnection of multi-terminal devices such as the disease prediction platform Web, mobile APP, and portable bracelets, thereby realizing long-term cardiovascular disease risk monitoring and helping people get rid of the troubles of cardiovascular diseases.

In this paper, considering that the platform needs to demonstrate good interpretability of the predicted results in the future, we only focused on traditional machine methods. However, deep learning methods have been demonstrated as a useful way for disease prediction in many works[9,44,45]. In future work, we aim to explore and assess the performance of GBDT+LR compared with the deep learning methods based on the research on the interpretability of deep learning.

For data citations of datasets uploaded to e.g. *figshare*, please use the `howpublished` option in the bib entry to specify the platform and the link, as in the `Hao:gidmaps:2014` example in the sample bibliography file.

Data availability

All data supporting the findings of this study are available within the paper.

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Author contributions

All authors significantly contributed to the scientific study and writing. Zengxiao Chi contributed to the overall idea, model formulation and analysisand was responsible for the writing of the manuscript; Li Liu was responsible for program implementation, including model construction and web system development. Lin Shi contributed to the process of refining research ideas. LiqinYi assisted in refining research ideas and polishing the entire article. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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